The Thermal Lens Revisited: Emergence of Microstructure, Scarcity-Induced Curvature, and Hot-Spot Regularization

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In honor of Andrej Cherkaev and Graeme Milton

We celebrate forty years of progress on this foundational problem and motivate the inevitable emergence of microstructure with new insights. We then present what are, to our knowledge, the first numerical results for the volume-fraction-constrained heat lens; Velo (2000) gave the first for volume-constrained gradient-field design in electrostatics. These results also address an engineering concern: hot spots, both throughout the domain and across the window itself.

AI-assisted research and numerics, run as a parallel experiment to the work itself.

The thermal lens: forty years

	Reference	What it extends
1985/88	A. Cherkaev, L. Gibiansky, K. Lurie, "Optimum focusing of heat flux by means of a inhomogeneous heat conducting medium," Report 305, Technical University of Denmark (1985); published (L. Gibiansky, K. Lurie, A. Cherkaev) as "Optimum focusing of a heat flux by a nonuniform heat-conducting medium (the 'heat lens' problem)," *Zh. Tekh. Fiz.* 58 (1988) 67–74	Poses the problem and solves it via the G-closure
1992	K. A. Lurie, "Direct solution of an optimal layout problem for isotropic and anisotropic heat conductors on a plane," *JOTA* 72(3) (1992) 553–575	Method: direct relaxation
1995	R. Adams, "Direct solution of an optimal layout problem for isotropic heat conductors with a volume fraction constraint," *JOTA* 85 (1995) 545–561	Constraint: limited good conductor
2000	A. P. Velo, "Optimal Design of Gradient Fields with Applications to Electrostatics," Ph.D. dissertation, Worcester Polytechnic Institute (2000)	Volume-fraction constraint in gradient-field design (electrostatics), with new numerical results: the closest prior problem
2001	R. Lipton, "Optimal inequalities for gradients of solutions of elliptic equations occurring in two-phase heat conductors," *SIAM J. Math. Anal.* 32(5) (2001) 1081–1093	Bounds on gradient size
2023	A. Cherkaev, E. Cherkaev, K. Lurie, "Optimal structures for focusing and energy accumulation: mathematical models and intuition," *Proc. R. Soc. A* 479 (2023) 20220342	Uncertain inputs and space–time materials
2026	R. Adams, K. Lurie, W. Sanguinet, in preparation	Emergence of microstructure, scarcity, the volume-fraction field

We later found the same nonlocal form, with the opposite sign, in block copolymers: T. Ohta, K. Kawasaki, "Equilibrium morphology of block copolymer melts," *Macromolecules* 19 (1986) 2621; Y. Nishiura, I. Ohnishi, "Some mathematical aspects of the micro-phase separation in diblock copolymers," *Physica D* 84 (1995) 31–39.

The stage

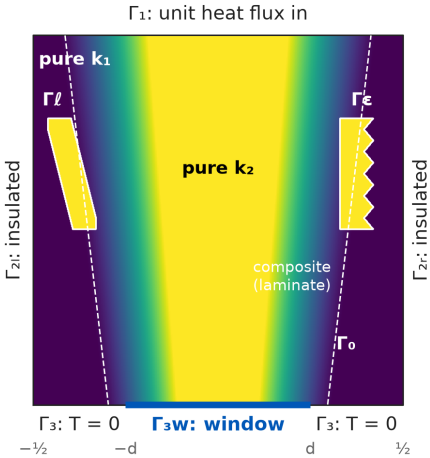


Plate and boundaries. $S = [-\frac{1}{2}, \frac{1}{2}]^2$, the unit square; Γ_1 top, unit flux in; $\Gamma_{2\ell}, \Gamma_{2r}$ sides, insulated; Γ_3 bottom, $T = 0$; window $\Gamma_{3w} = [-d, d]$, $d = \frac{1}{4}$.

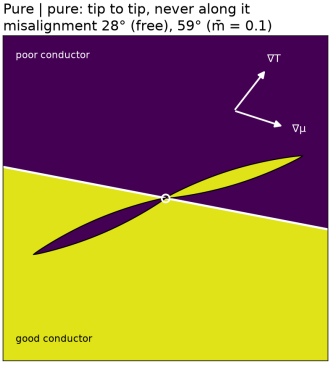
State. Two conductors $0 < k_1 < k_2$ ($k_2/k_1 = 10$); temperature T, flux $\boldsymbol{q} = -\boldsymbol{K} \cdot \boldsymbol{\nabla} T$; conjugate μ. Objective: heat through the window, $I = \int_{\Gamma_{3w}} \boldsymbol{e}_2 \cdot \boldsymbol{K} \cdot \boldsymbol{\nabla} T$; $J = I$ is the captured fraction (total heat in $= 1$).

Design. Pure: $\boldsymbol{K} = k_1 \chi_1 + k_2 \chi_2$. Relaxed: a rank-1 laminate $\boldsymbol{K}_0$ with fraction m of k_2 and layer angle φ; χ (on later slides) $= \angle(\boldsymbol{\nabla}\mu, \boldsymbol{\nabla} T)$, $p = (k_2 - k_1)/(k_2 + k_1)$.

Constraint and regularizers. $\overline{m}$: target mean of m, multiplier κ; volume-fraction potential ω, $\nabla^2 \omega = m - \overline{m}$; ε weights $\int |\boldsymbol{\nabla}\omega|^2$ and makes the price local, $\kappa = \kappa_0 - \varepsilon\omega$.

Interfaces $\Gamma_* = \Gamma_0 \cup \Gamma_\ell \cup \Gamma_\varepsilon$: Γ_0, macroscopic edge of a deployment regime (e.g. of the composite); Γ_ℓ, mesoscopic pure | pure boundary; Γ_ε, microscopic pure | pure boundaries.

Kissing inclusions: no smooth romance



Γ_ℓ: a pure | pure interface; $\pm$: its two sides.

A thin strip Λ with normal $\boldsymbol{n}$ (the strongest inclusion) must not pay:

$$\frac{\Delta J}{|\Lambda|} \to \boldsymbol{\nabla}\mu \cdot \Delta\boldsymbol{K}_0 \cdot \boldsymbol{\nabla} T - \frac{(\boldsymbol{n} \cdot \Delta\boldsymbol{K}_0 \cdot \boldsymbol{\nabla}\mu)\,(\boldsymbol{n} \cdot \Delta\boldsymbol{K}_0 \cdot \boldsymbol{\nabla} T)}{\boldsymbol{n} \cdot (\boldsymbol{K}_0 + \Delta\boldsymbol{K}_0) \cdot \boldsymbol{n}} \leq 0 \qquad \text{for every } \boldsymbol{n}$$

On a pure | pure interface, two conditions on one curve:

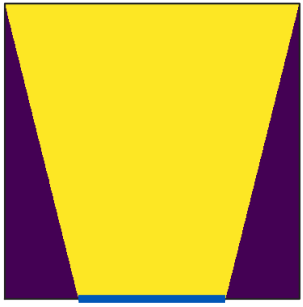
- Weierstrass–Erdmann: $\boldsymbol{\nabla}\mu^+ \cdot \boldsymbol{\nabla} T^- = 0$
- kissing: the tangent lies along the bisector of $\boldsymbol{\nabla}\mu$ and $\boldsymbol{\nabla} T$

Over-determined: smooth pure | pure interfaces are not optimal and the designs keep refining (Lurie 2026); in a relaxed optimum, Γ_ℓ is generically empty: no pure | pure contact.

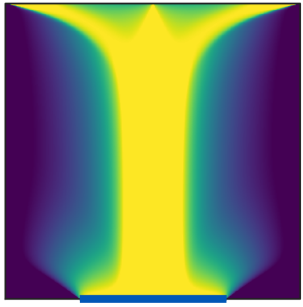
From a naive funnel to microstructure

$J = -\int_{\Gamma_1} \mu + \int_S \nabla\mu \cdot \boldsymbol{K} \cdot \nabla T, \quad \boldsymbol{K} = k_1\chi_1 + k_2\chi_2$

naive funnel: J = 0.9019



relaxed optimum: J = 0.9372

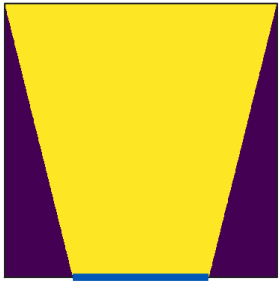


$k_1 = 1$ $k_2 = 10$ Unconstrained, $h = 0.0005$, needle deployments. The funnel's walls fail the Smooth Deployment Law; tongues of both phases grow from them into the relaxed composite zone. Cell flips alone reach $J = 0.9324$.

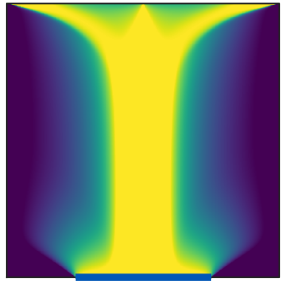
From a naive funnel to microstructure

$J = -\int_{\Gamma_1} \mu + \int_S \nabla\mu \cdot \boldsymbol{K} \cdot \nabla T, \quad \boldsymbol{K} = k_1\chi_1 + k_2\chi_2$

naive funnel: J = 0.9009



relaxed optimum: J = 0.9372



$k_1 = 1$ $k_2 = 10$ **Projector backup** (coarser tongues). Unconstrained, $h = 0.001$, needle deployments.

The funnel's walls fail the Smooth Deployment Law; tongues of both phases grow from them into the relaxed composite zone. Cell flips alone reach $J = 0.9313$.

Pure phases, with needle deployments

$J = -\int_{\Gamma_1} \mu + \int_S \nabla\mu \cdot \boldsymbol{K} \cdot \nabla T - \kappa \int_S (\chi_2 - \overline{m}), \quad \boldsymbol{K} = k_1\chi_1 + k_2\chi_2$

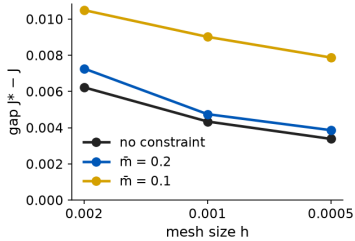
h = 0.002 (578k cells)

no constraint: J = 0.9310 | $\tilde{m}$ = 0.2: J = 0.9196 | $\tilde{m}$ = 0.1: J = 0.8761

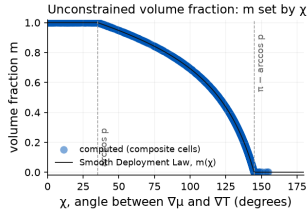


$k_1 = 1$ $k_2 = 10$ Unconstrained: $\kappa = 0$. Starts: the naive funnel; under a budget, a smooth trumpet over the window holding $\overline{m}$. Refinement adds fronds; J rises, still below the relaxed optimum.

Refinement pays



$\overline{m}$	h	needle deployments period	band
0.2	0.002	0.0140	0.025
0.2	0.001	0.0100	0.023
0.2	0.0005	0.0049	0.017
0.1	0.002	0.0140	0.026
0.1	0.001	0.0077	0.026
0.1	0.0005	0.0051	0.023

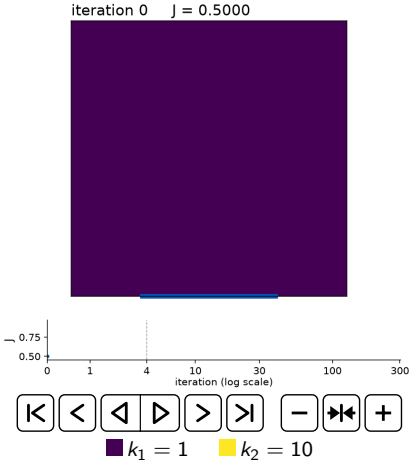


- From $h = 0.002$ to 0.0005 the gap shrinks by 46% (no constraint), 47% ($\overline{m} = 0.2$), 25% ($\overline{m} = 0.1$)
- Period falls 2.7–2.9× over two halvings; the band changes little
- What the tongues approach (right): unconstrained volume fraction, m set by χ alone

J^*: the relaxed (rank-1 laminate) optimum at the same budget. Funnel / trumpet starts. Period: good plus next poor stripe; band: run of stripes (< 0.02) on arcs about the window center.

$J = -\int_{\Gamma_1} \mu + \int_S \nabla\mu \cdot \boldsymbol{K}_0 \cdot \nabla T, \quad \boldsymbol{K}_0 \in G$

1985, revisited: the same scheme, run longer



This laptop is more than ten thousand times as powerful as the 1985 university machines.

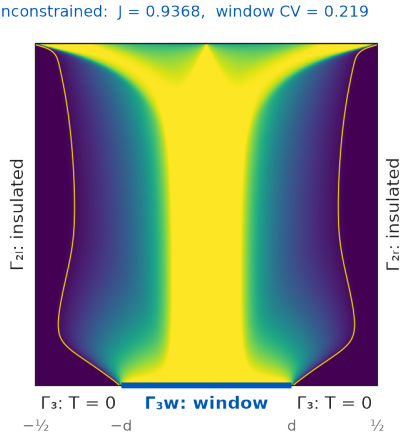
What running longer reveals

- Undamped, the scheme never settles: it flips between two states and slowly degrades ($J \approx 0.904$)
- Damped, from the 1985 design, it settles by iteration $\sim$20: $J = 0.9266 \rightarrow 0.9351$
- The gain comes both from how the material is deployed and from more good conductor: $\overline{m}$ rises $0.448 \rightarrow 0.468$ ($+4.5\%$)
- 1985 geometry (window 0.45 of the width), in J: published 0.85; our first iterate 0.906; settled 0.918
- The settled design shows what coarse grids could not: a graded laminate band between pure zones: across it, m runs along the Smooth Deployment Law from one pure end to the other

$h = 0.003$. m shown averaged over 3 cells: the raw fields carry cell-scale texture (± 0.03 in m) that changes neither J nor delivery.

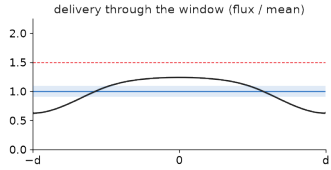
1995: a budget for the good conductor

$J = -\int_{\Gamma_1} \mu + \int_S \nabla\mu \cdot \boldsymbol{\kappa}_0 \cdot \nabla T - \kappa \int_S (m - \overline{m}), \quad \kappa = \text{const}$



The budget $\overline{m}$ shrinks frame to frame, from unconstrained to 0.03; the price κ that enforces it rises from 0 to 3.07.

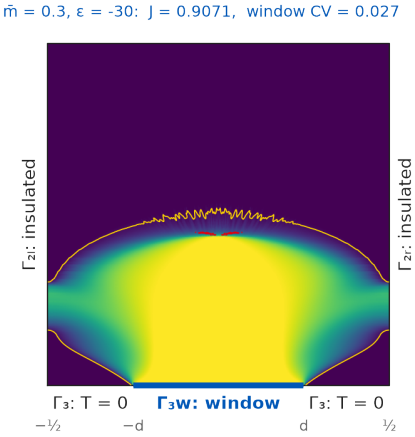
Gold: Γ_0, the edge of the composite. Red: hot spots, flux above 1.5× the mean delivered through the window.



$k_1 = 1$ $k_2 = 10$ Relaxed rank-1, $h = 0.001$, 1000 iterations each; m averaged over 3 cells.

2026: a local price for material

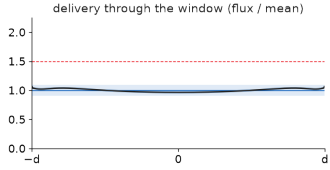
$$J = -\int_{\Gamma_1}\mu + \int_S \nabla\mu\cdot\boldsymbol{\kappa}_0\cdot\nabla T - \int_S \kappa\left[m-\overline{m}-\nabla^2\omega\right] - \tfrac{\varepsilon}{2}\int_S|\nabla\omega|^2$$



The same budgets as 1995, now with the local price $\kappa = \kappa_0 - \varepsilon\,\omega$, ε tuned for each budget: $|\varepsilon|\,\overline{m}^2 \approx 2.5$.

Gold: Γ_0. Red: hot spots, flux above 1.5× the mean delivered through the window.

Fine texture at the crown of Γ_0 changes J by $\sim 10^{-4}$ and leaves delivery unchanged; the lay-out sets delivery.





$k_1 = 1$ $k_2 = 10$ $h = 0.001$. At $\overline{m} = 0.03$, $\varepsilon = -1000$ reaches only $|\varepsilon|\,\overline{m}^2 = 0.9$.

$$J = -\int_{\Gamma_1} \mu + \int_S \nabla\mu \cdot \boldsymbol{K}_0 \cdot \nabla T - \int_S \kappa\,[m - \overline{m} - \nabla^2\omega] - \tfrac{\varepsilon}{2}\int_S |\nabla\omega|^2$$

The volume-fraction field

A potential for the volume fraction,

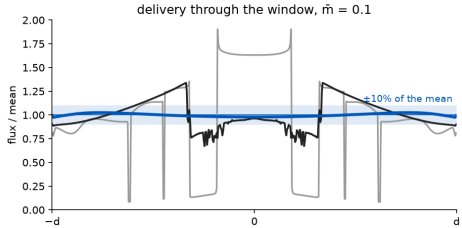
$$\nabla^2\omega = m - \overline{m}, \qquad \partial_n\omega = 0,$$

adds $-\frac{\varepsilon}{2}\int |\nabla\omega|^2$ to J and makes the price of material local:

$$\kappa(\boldsymbol{x}) = \kappa_0 - \varepsilon\,\omega(\boldsymbol{x}).$$

- $\varepsilon < 0$ rewards concentration
- it settles the iteration about 10× faster: it removes the flat directions

$\int |\nabla\omega|^2 = \int (m - \overline{m})(-\nabla^2)^{-1}(m - \overline{m})$: formally the nonlocal term of Ohta and Kawasaki[1], in nearly the form of Nishiura and Ohnishi[2]; $\varepsilon < 0$ gives it the attractive sign.



At $\overline{m} = 0.1$:

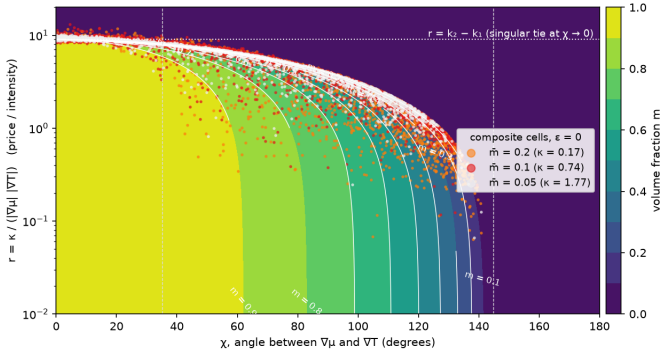
	pure	laminate $\varepsilon = 0$	laminate $\varepsilon = -300$
J	0.8776	0.8866	0.8575
window CV	0.46	0.15	0.016
window min / mean	0.08	0.67	0.98
domain CV of |flux|	0.76	0.81	0.49

Against the best pure design (needle deployments) the ε design gives up 2.3% of J: the price of even delivery.
$h = 0.001$; domain CV inside the layers (Velo 2000).

[1] T. Ohta, K. Kawasaki, *Macromolecules* 19 (1986) 2621. [2] Y. Nishiura, I. Ohnishi, *Physica D* 84 (1995) 31–39.

$J = -\int_{\Gamma_1} \mu + \int_S \nabla\mu \cdot \boldsymbol{K}_0 \cdot \nabla T - \kappa \int_S (m - \overline{m})$

The Smooth Deployment Law under scarcity



Under a budget, $m = m(\chi, r)$ with $r = \kappa/(|\nabla\mu|\,|\nabla T|)$; the budget doesn't change the law; it raises the price κ, which moves the cells higher on the same map; unconstrained is the bottom edge, $r \to 0$.

Inverting the law cell by cell*: the composite obeys it at the computed price everywhere except a thin collar on the axis above the window, the singular tie, where the law cannot choose m.

* Implied price: $r^* = \frac{k_2 - k_1}{2}\left[(1 + \cos\chi) - \frac{k_1 k_2\,(1 - \cos\chi)}{\left(k_2 - (k_2 - k_1)\,m\right)^2}\right]$, $\kappa^* = r^*\,|\nabla\mu|\,|\nabla T|$; $r^*(\chi, 0) = r_0$, $r^*(\chi, 1) = r_1$, $r^*(0, m) = k_2 - k_1$ (the singular tie). Checked in SymPy. $h = 0.001$, $\overline{m} = 0.1$, restarted with relaxation 0.005: off the tie, rms $|m - m(\chi, r)| = 0.007$; the tie collar is 10% of the composite.

Summary and directions

- The Smooth Deployment Law sets the graded laminate; scarcity turns it into a family indexed by price/intensity
- Smooth pure | pure interfaces fail the kissing test; jumps in the relaxed designs run along the layers, where the test is satisfied
- Composites over the best pure designs: $+0.4$–1.0% in J at equal material; at equal J, 8–9% less good conductor under scarcity, about half unconstrained
- The volume-fraction field consolidates the scarce designs into one dome wrapped in rank-1 laminate, evens window delivery, and lowers the domain flux CV; ε-tuned at $|\varepsilon|\,\overline{m}^2 \approx 2$–$3$ (this geometry)
- The limit: below $\overline{m} \approx 0.05$ no ε gives an even supply; the window starves

Next

- The dual problem: least good conductor for a required capture
- Prescribing the delivery profile and shadowing it back into the design
- Refine the structure of the local cost to be analytically consistent and see how far we can push hot spot reduction

Postscript: AI-assisted research

1. Using AI makes you feel like a genius. But, it also makes you feel dirty. You are not a dirty genius. Leave the keyboard behind for the pencil often. Leave both behind for a good think daily.
2. Don't treat it like a trusted colleague (an acquired delusion), you will spend a lot of time going in circles.
3. Don't get too enamored by the ease with which you can explore new ideas; you will lose yourself and find that you have done nothing substantive.
4. It really is remarkably good at writing code. Coding languages are much more structured than human languages. It has 50+ years of FOSS to train on too.
5. If you use it to check calculations, insist that it drive them through an auditable tool like SymPy. You could end up with garbage otherwise.
6. Your new job is to be the Critical Thinker in Chief; do not accept anything an AI tells you.
7. Pit multiple AIs against each other and tell them that you are doing so. They do a better job of checking each other's work.
8. Be explicit in your prompts. When you feel that you are being vague, remind the AI that it may have questions. This patches up vaguer prompts.
9. Leverage AI to generate your slides in LaTeX/Beamer. It is so quick at this. Just don't keep revising the slides until the day before the talk (like I have).

Backup

Backup: when can pure meet pure smoothly?

On Γ_ℓ (good k_2 on the $+$ side, poor k_1 on the $-$ side): T_t and the normal flux $q_n = k\,\partial_n T$ are continuous, and likewise for μ.

Kissing (the tangent along the bisector of $\nabla\mu$ and ∇T): $\nabla\mu$ is the mirror image of ∇T in the tangent, on both sides with the same factor $\lambda > 0$.

Weierstrass–Erdmann: $(k_1 - k_2)\,\nabla\mu^+ \cdot \nabla T^- + \kappa = 0 \implies \lambda\,(k_2 - k_1)\Big(T_t^2 - \dfrac{q_n^2}{k_1 k_2}\Big) = \kappa.$

$\kappa = 0$ (unconstrained volume fraction)

$q_n^2 = k_1 k_2\, T_t^2 \implies \cos\chi^+ = p, \quad \cos\chi^- = -p$

The good side sits exactly at $\arccos p$ and the poor side at $\pi - \arccos p$: the thresholds of the Smooth Deployment Law.

$\kappa = \kappa_0 - \varepsilon\,\omega$ (budget, local price)

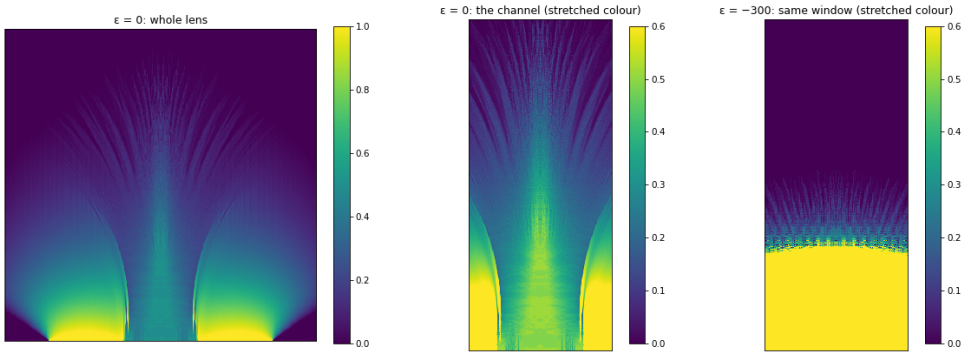
$r^+ = r_1(\chi^+), \qquad r^- = r_0(\chi^-)$

with $r = \kappa/(|\nabla\mu|\,|\nabla T|)$ on each side. Each side sits exactly on the edge of its own pure regime ($m = 1$ and $m = 0$), at its own price/intensity.

Either way, a smooth Γ_ℓ is admissible only as a composite band of zero width: the two envelopes Γ_0 coincide. Off that, one side falls into the composite window and the band opens. In the limit $q_n \to 0$: $\chi \to 0$ on both sides and $r \to k_2 - k_1$, the tie of the near-singular column.

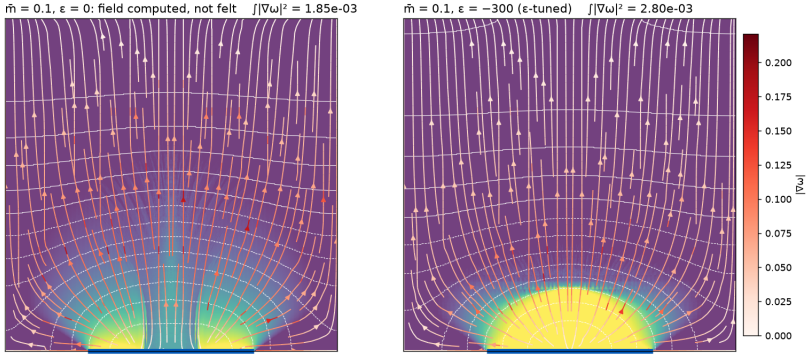
$p = \frac{k_2 - k_1}{k_2 + k_1}$; $r_1, r_0 = \frac{k_2 - k_1}{2}\big[(1 + \cos\chi) - \frac{k_2}{k_1}(1 - \cos\chi)\big]$, same with $\frac{k_1}{k_2}$. With the local price, ω and $\nabla\omega$ are continuous across Γ_ℓ, so κ enters pointwise. SymPy: *Calculations.ipynb*.

Backup: the raw fields carry cell-scale texture



Raw m at $\overline{m} = 0.1$ (centre and right: colour range stretched to 0–0.6). The feathers are 1–3 cells wide at every mesh (half-width $0.0045 \to 0.0018 \to 0.0009$ for $h = 0.003 \to 0.001 \to 0.0005$) and only ± 0.02–0.03 in m. Smoothing them changes J by 10^{-4} and leaves window delivery unchanged, so the movies show m averaged over 3 cells. The channel's trunk, taper and field-aligned layering are far larger and survive the averaging.

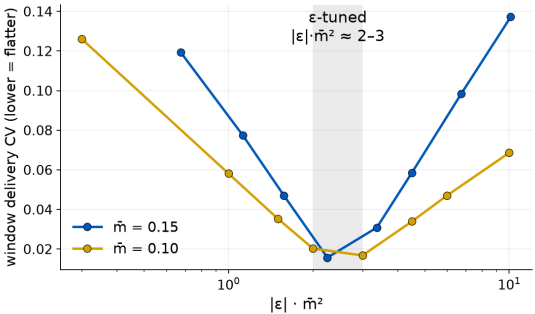
Backup: what ω is



ω is a gravitational potential: $\nabla^2\omega = m - \overline{m}$ is Poisson's equation $\nabla^2\Phi = 4\pi G\rho$, with the material as the mass (its mean subtracted, as a uniform background). ω is lowest where the good conductor is densest (in its pure core, with ε tuned); the field lines of $\boldsymbol{\nabla}\omega$ (red) run out to the empty region, and the pull $-\boldsymbol{\nabla}\omega$ points back in.

$\varepsilon < 0$ rewards $\int |\boldsymbol{\nabla}\omega|^2$, a deeper binding, so material is drawn together into one place, as under gravity: $\int |\boldsymbol{\nabla}\omega|^2$ rises from 1.85×10^{-3} to 2.80×10^{-3} with ε tuned.

Backup: tuning ε



Settled rank-1 runs, $h = 0.003$. The constant is specific to this geometry and contrast. Mesh check at $\varepsilon = -300$, $\overline{m} = 0.1$: J 0.8551–0.8579, window CV 0.016–0.017, $h =$ 0.003–0.00075.

- It smooths the flux over the whole plate (lower domain CV); with ε tuned, the window profile nearly mirrors the domain
- Pushed further, ε makes the forbidden kiss possible (pure meets pure at $\varepsilon = -1000$; at $\overline{m} = 0.03$ already at $|\varepsilon|\,\overline{m}^2 \approx 0.9$): such designs are likely inadmissible

Backup: the Cherkaev–Gibiansky–Lurie (1985) problem on its own geometry

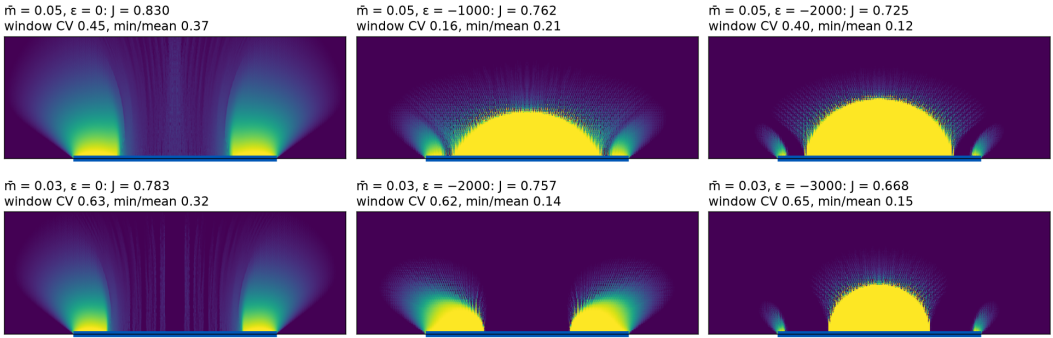
Square of half-width 5, window 4.5 long, unit flux in at the top; I = heat through the window (I_0: isotropic body).

	I	I/I_0
isotropic body	4.50	1.00
published 1985 (Fig. 3)	8.5	1.89
same scheme, our iteration 1	9.06	2.01
same scheme, iterations 3–20	8.86–9.09 (oscillates)	
damped, settled	9.18	2.04

Our runs reproduce $I_0 = 4.5$ exactly. On this geometry the 1985 stopping rule never fires within 20 undamped iterations; the published value lies below even our first iterate, plausibly from the 1985 grid.

$h = 0.03$ (258 520 cells), $u_+/u_- = 10$.

Backup: the scarce end, where no ε gives an even supply



At $\overline{m} = 0.05$ the strongest ε forms a dome and lowers the window CV, but leaves cold bands: the weakest point still gets only 21% of the mean. At $\overline{m} = 0.03$ nothing tried lifts it above 15%. Compare 98% with ε tuned for $\overline{m} = 0.1$.

$\varepsilon = 0$: $h = 0.001$; $\varepsilon \neq 0$: $h = 0.003$. Lower part of the plate shown.

Backup: the model behind the numbers

Solver

- DOLFINx 0.11 (FEniCSx) on PETSc; meshes from gmsh, uniform, built on half the plate and mirrored (reflection symmetry), cached
- Temperature T and conjugate μ: P1 Lagrange; conjugate gradients with hypre AMG, relative tolerance 10^{-10}
- Design: one conductivity tensor $\boldsymbol{K}_0$ per cell (DG0), with fraction m and layer angle φ
- Objective: reaction flux through the window (exact for uniform flux), as a fraction of the incoming heat
- Schemes: the 1985 update; Adams 1995; ALS 2026 rank-1 laminates; pure phases (0/1); restart from any saved design
- Runs: TOML scenarios $\rightarrow$ batch CLI; every run records a replay; an interactive notebook with live controls

Knobs

- Physics: contrast k_2/k_1; plate size; window width and position; lateral boundaries (insulated or held); maximize or minimize
- Material: target $\overline{m}$ (multiplier κ solved each step) or free; laminate laws (rank-1, three deployments, straight or curved laminates); pure, Bruggeman and Hashin–Shtrikman alternatives
- Regularizer: ε, the volume-fraction field (local price $\kappa_0 - \varepsilon\omega$, spectral Neumann solve)
- Iteration: damping (relaxation), switch cap for pure designs, curvature damping, projected gradient, tolerance
- Filters and smoothers: fraction filter (radius); budget window (one multiplier per local tile)
- Diagnostics: window delivery statistics; domain CV inside the layers; interface, kissing and Weierstrass–Erdmann audits; convergence of J